

3.4 BANDWIDTH CONTROLS OPTIMIZATION ACCESS TO HIGH PRECISION

The construction’s nearly width-independent relative bandwidth $\lambda = \gamma h$ requires $\gamma = \Theta(W)$. Even though larger widths admit smaller λ at comparable construction error (Figure 2b), small slopes can delay readout optimization. We quantify this in the dictionary of Section 3.1, replacing the direct normalized least-squares solve with GD while keeping the features fixed.

Theorem 3.2 (Slope-dependent output-error lower bound). *For the fixed-center construction of Section 3.1 with common slope $\gamma > 0$ and nonzero sampled target, readout GD from zero with $\eta = 1/(2\mu_1)$ satisfies*

$$\frac{\|f - q_n\|_{2,Z}^2}{\|f\|_{2,Z}^2} \geq \sum_i p_i(\gamma) \left(1 - \frac{\bar{\rho}_i(\gamma)}{2}\right)^{2n}, \quad n \geq 0. \quad (1)$$

Here $\|\cdot\|_{2,Z}$ is the training-sample RMS norm, μ_i are the descending readout-kernel eigenvalues, p_i is the fraction of target energy in eigenvector i , and $\bar{\rho}_i(\gamma) \in [0, 1]$ bounds μ_i/μ_1 from above. The explicit gamma-dependent rate bounds and finite-geometry allowances are given in Appendix A.

Proof sketch. Gamma enters the feature correlations through $d \coth(\gamma d)$, where d is the separation between training points. Controlling the finite-center corrections gives the relative-rate bounds; a standard spectral argument then yields the output-error inequality.

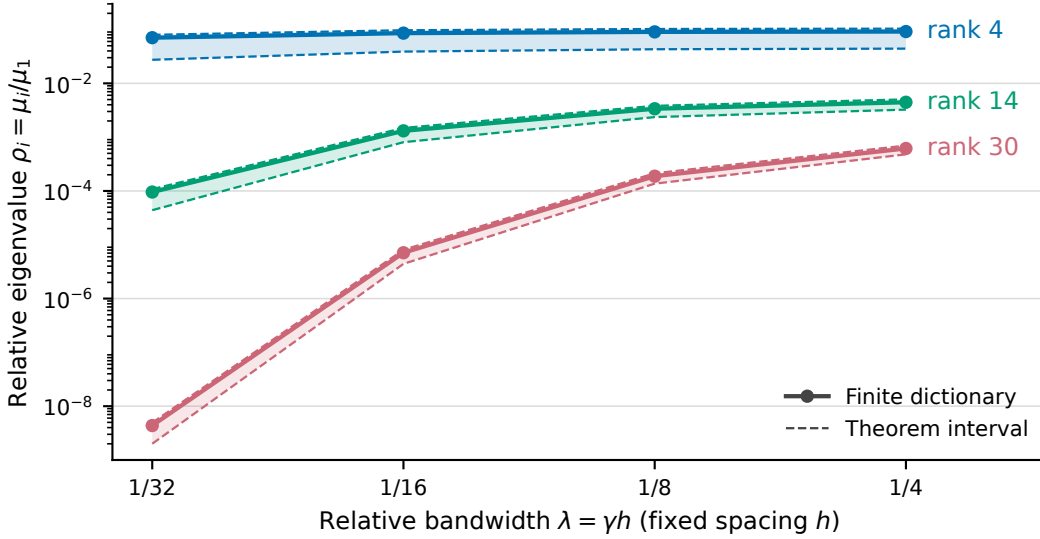


Figure 3: **Slope changes relative rates by very different amounts.** Finite-kernel ratios (solid, dots) and theorem intervals (dashed boundaries, shading) for three illustrative ordered ranks, at fixed width 512. Four bandwidths are evaluated; connecting lines are guides. Rank 30 carries 2.6% and 2.5% of target energy at the two endpoints, while its relative rate increases by 1.4×10^5 . Ordered rank does not identify the same eigenvector across slopes. All figures use the target and geometry in Appendix B.

Increasing λ from $1/32$ to $1/4$ raises the thirtieth relative eigenvalue by 1.4×10^5 , versus only 1.3 for the fourth (Figure 3); the thirtieth mode carries about 2.5% of target energy at both endpoints.

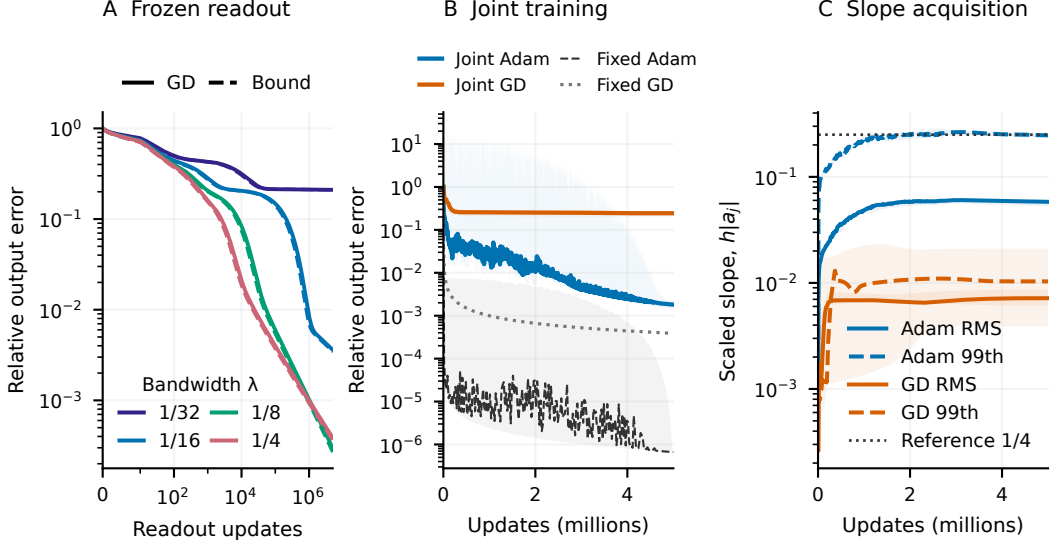


Figure 4: **From readout delay to feature acquisition.** **A:** executed frozen-readout GD errors (solid) and the square roots of the theorem’s lower bounds (dashed). **B:** joint Adam/GD errors, with frozen uniform $\lambda = 1/4$ references over the same five-million-update budget. **C:** RMS and 99th-percentile scaled slopes $h|a_j|$ from B’s joint runs. The reference bandwidth is a supplied geometry, not a necessary threshold for heterogeneous features. Joint curves summarize five paired seeds; shading retains seed variation and recorded within-bin extrema. The full horizon is shown and cosine schedules span that horizon. Appendix B specifies the error measure and selection protocol. The theorem applies to A; B–C are empirical.

At $\lambda = 1/32$, a direct readout fit attains 6.2×10^{-10} dense-grid relative error, yet five million GD updates retain training error 0.210 against the theorem’s lower bound 0.2098 (Figure 4A). At $\lambda = 1/4$, GD reaches 3.88×10^{-4} over the same budget. Across the four bandwidths, the bound remains at least 75% of executed error throughout training: small slopes delay accuracy already available in the dictionary.

Over the same budget, joint Adam reaches median dense-grid relative error 1.81×10^{-3} , versus 6.62×10^{-7} on the supplied $\lambda = 1/4$ features, despite partial slope acquisition (Figure 4B–C). This motivates the next analysis: training must acquire useful slopes and centers together. The construction supplies both the feature geometry and the readout directly.